

[Eccles Building, U.S. Federal Reserve
Board of Governors](#)



ECON 100B: Macroeconomics

Class 19: Monetary Policy & the Phillips Curve [2/2]
(Jones Chapter 12)

Ryan D. Edwards
April 7, 2026

Agenda

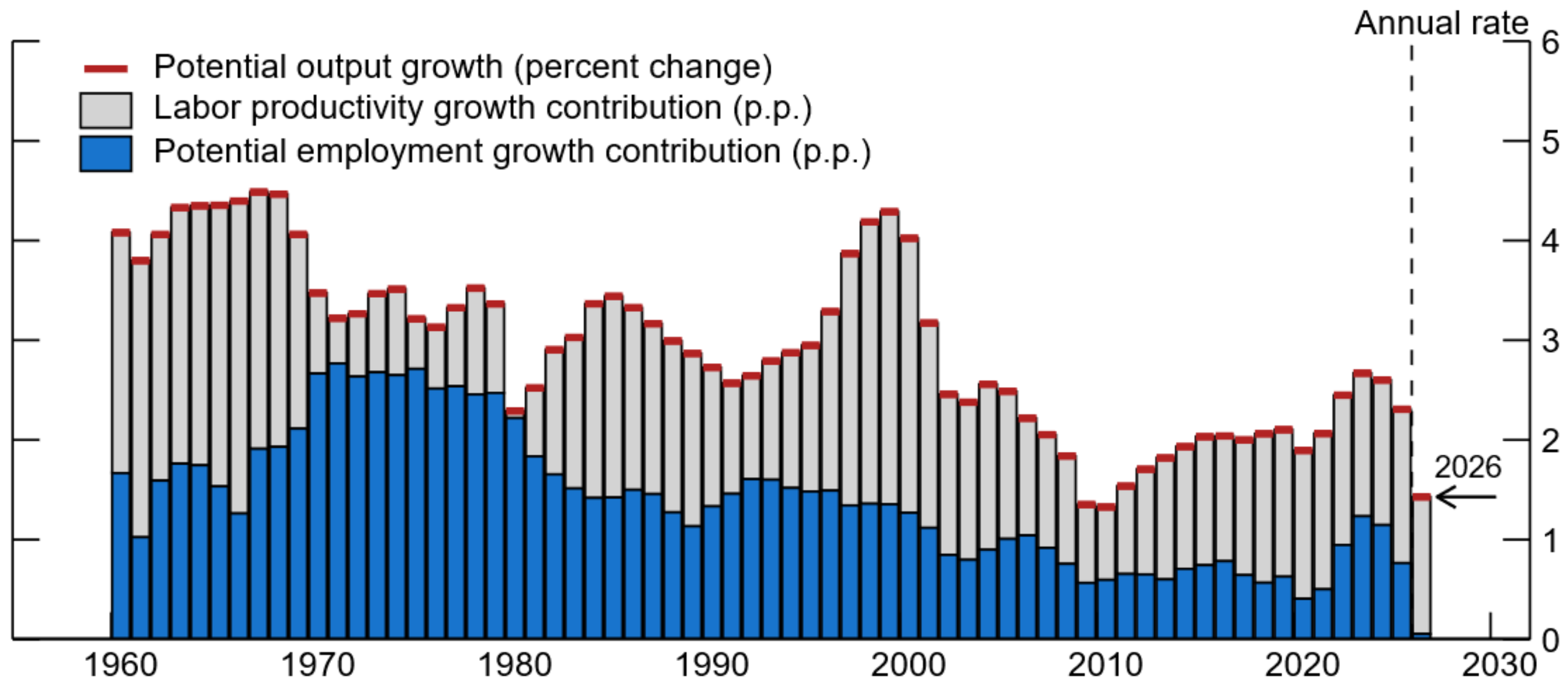
- Finish Chapter 12: Monetary policy and the Phillips curve

Announcements

- **Problem Set 3** is available, due on **Friday, April 17**
- **Problem Set 4** will be due on **Friday, May 1**
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in **Pimentel 1**
 - We will run 90 minutes from about 3:00-4:30PM
 - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**
(I think it's kind of near Caffe Strada on Bancroft)

Current events

- Murray and Vidangos (4/2/26) “[Labor force growth, breakeven employment, and potential GDP growth](#),” FEDS Notes
- Recall the Solow model: steady-state Y grows at rate n plus any growth in productivity $g[A]$



More current events

- Natasha Sarin (4/6/26) “[This Is Starting to Look Like a Slow-Motion Bank Run](#),” NYT
 - “Private Credit” lenders direct about \$2 trillion of funds toward usually riskier borrowers
 - Bank loans are about 12 times larger
- Pritcher (3/29/26) “[Private Credit’s Exposure to Ailing Software Industry Is Bigger Than Advertised](#),” WSJ
- Andriotis (4/2/26) “[The Wealthy Investors That Powered Private Credit Are Rushing for the Exits](#),” WSJ
- The concern:

Those exposed to the private credit industry might be large relative to the market. See for example [AIG](#) in the 2008 crisis, bailed out by the Federal Reserve

Learning objectives for Chapter 12

- The full short-run model consists of
 - A downward-sloping **IS curve**
 - A horizontal **MP curve** at the interest rate R chosen by the central bank
 - The **Phillips curve**, an upward-sloping relationship between $\Delta\pi$ and $\tilde{Y}$
- External price shocks can raise or lower the Phillips curve
- To reduce inflation through the Phillips curve, the central bank must cause a (short?) recession with $\tilde{Y} < 0$
- The Great Inflation disaster of the 1970s in the U.S. was partly due to monetary policy that was too “loose,” meaning R was too low

These are **NOT** learning objectives for ECON 100B

- How monetary policy is actually conducted by the Fed and other central banks
 - The “nuts and bolts” have been in flux over the decades
 - But the bottom line is the same:

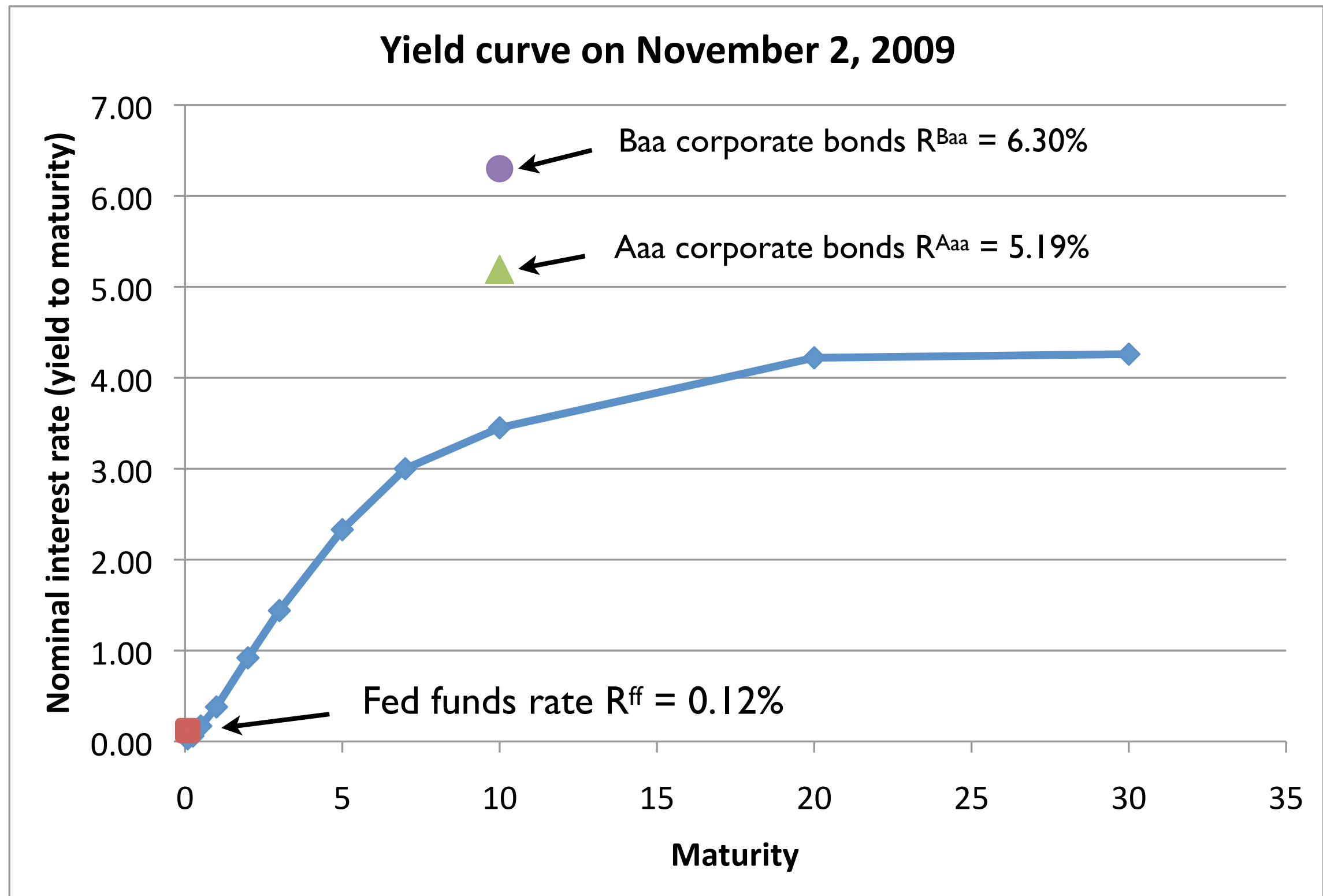
The Fed and other central banks **set the short-run real interest rate**
(with some constraints)

- The Fed’s balance sheet, quantitative easing

The federal funds rate R^{ff} and other interest rates

- The federal funds rate R^{ff} is the rate that banks charge each other on overnight loans
- It is a **short-term** interest rate
- The Fed can effectively set the value of R^{ff} because banks could also borrow from the Fed through the discount window
- Businesses that invest in capital typically require loans with much longer terms or “maturities”
- Banks typically charge **term premiums** on longer loans. They also charge **risk premiums** when the borrower is riskier

The Treasury yield curve shows term premiums.
We can also see risk premiums in 10-year corporate bonds



Technically the Fed affects *nominal interest rates*. What about the real interest rate?

- We know that the nominal interest rate, i_t , is related to the real interest rate, R_t , through the Fisher equation:
- Does inflation change? Sure, but *not in the short run*
- This will be a key assumption of ours, and it's basically right: inflation will not change a lot over short periods of time
- So changes in the nominal federal funds rate will typically produce changes in the real federal funds rate, R^{ff}

Technically the Fed sets *nominal interest rates*. What about the real interest rate?

- The nominal interest rate i is related to the real interest rate R and inflation through the Fisher equation:

$$R = i - \pi$$

- Does inflation change? Sure, but if inflation is under control, it doesn't change much in the short run
- Changes in the nominal federal funds rate will typically produce changes in the real federal funds rate, R^{ff}

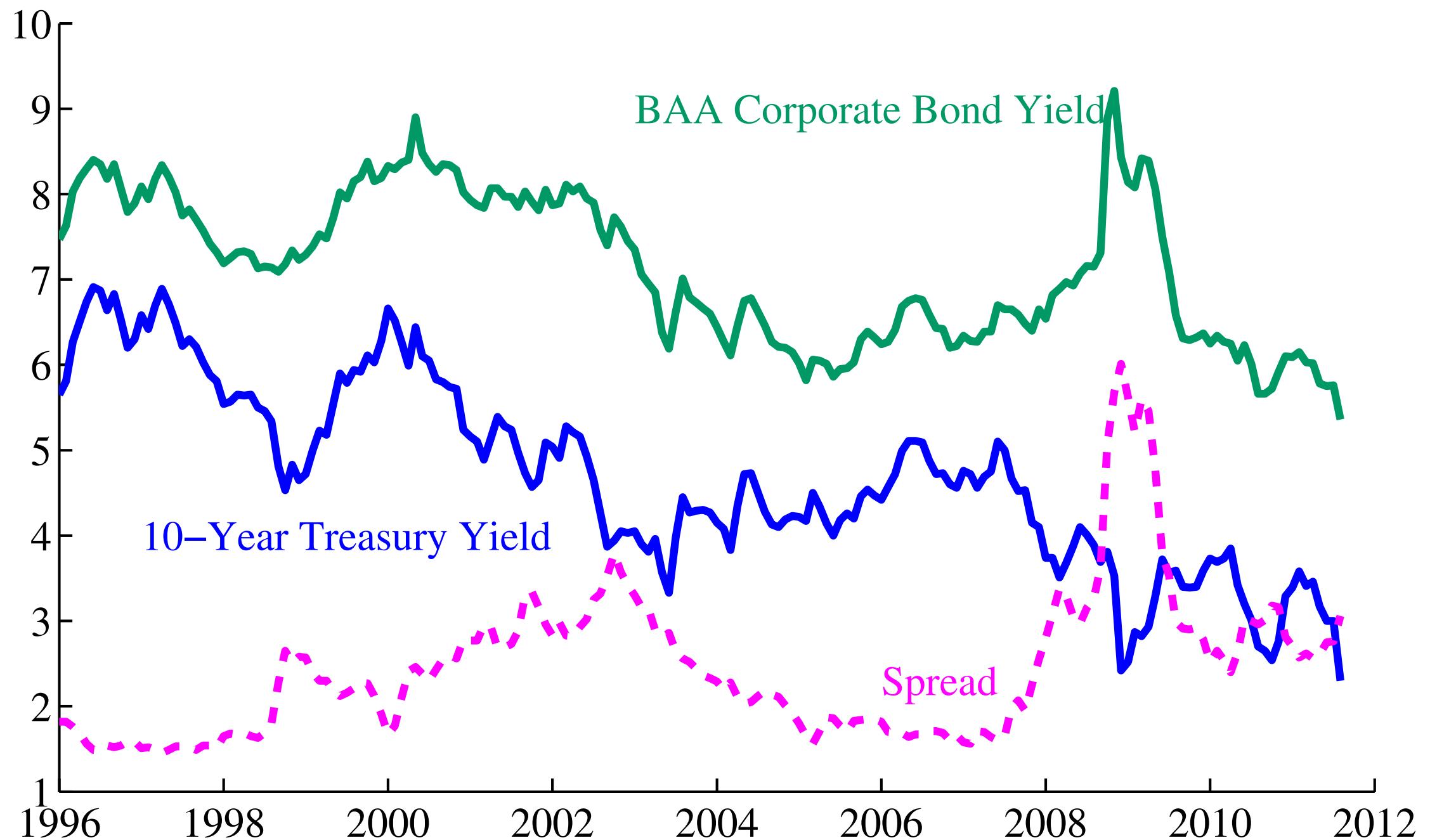
The Fed sets the federal funds rate, R^{ff} , but markets set the risk premium, $\bar{p}$

- The Fed can choose R^{ff}
- But then financial markets decide what risk premium $\bar{p}$ they need to charge on business loans
- (Term premiums matter, but they are more predictable than risk premiums, which can really wreak havoc)
- In our model, we'll assume there is one “average” interest rate R that is important for businesses in the IS curve

$$R = R^{ff} + \bar{p}$$

- During normal times, we'll assume that $\bar{p} = 0$

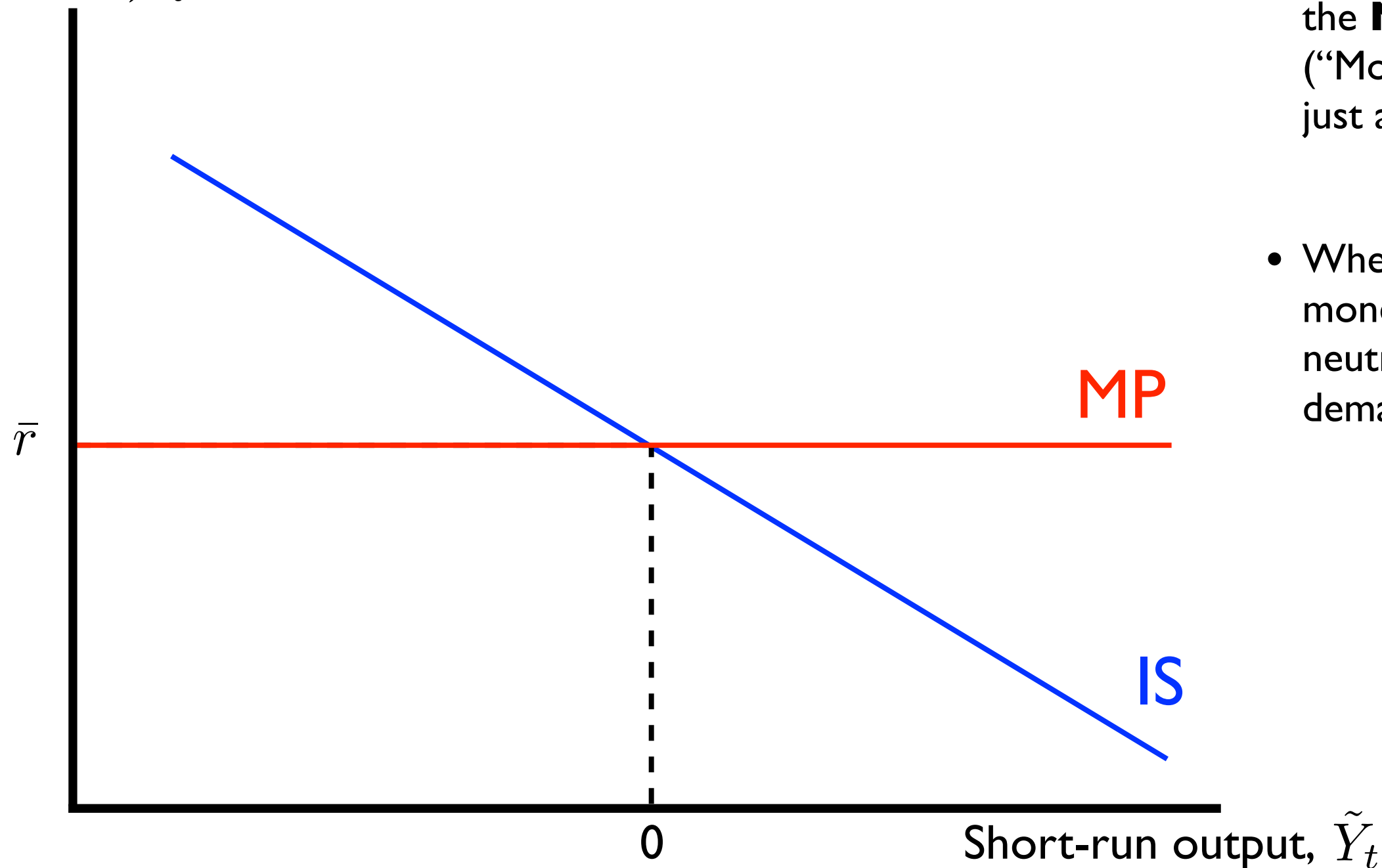
Times have notably *not* been normal



The “**spread**” between Baa corporate bonds and the 10-year Treasury, which is the **risk premium** $\bar{p}$, increased dramatically in 2008

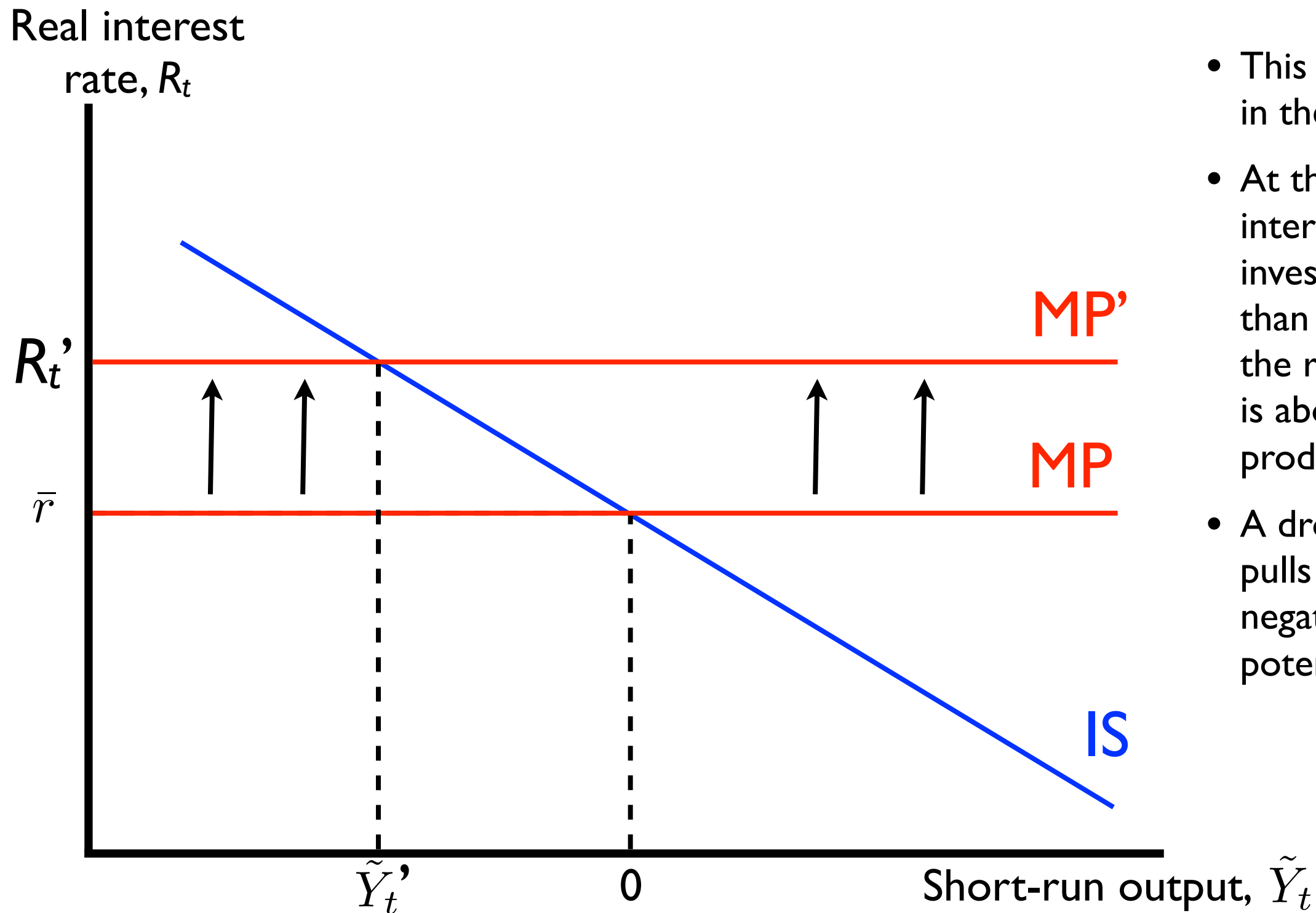
If the Fed can determine the real interest rate in the short run, then here's our new short-run model diagram:

Real interest
rate, R_t



- The Fed can set the real interest rate, so the **MP Curve** (“Monetary Policy”) is just a horizontal line
- When $R = \bar{r}$, monetary policy is neutral. Without any demand shocks, $\tilde{Y} = 0$

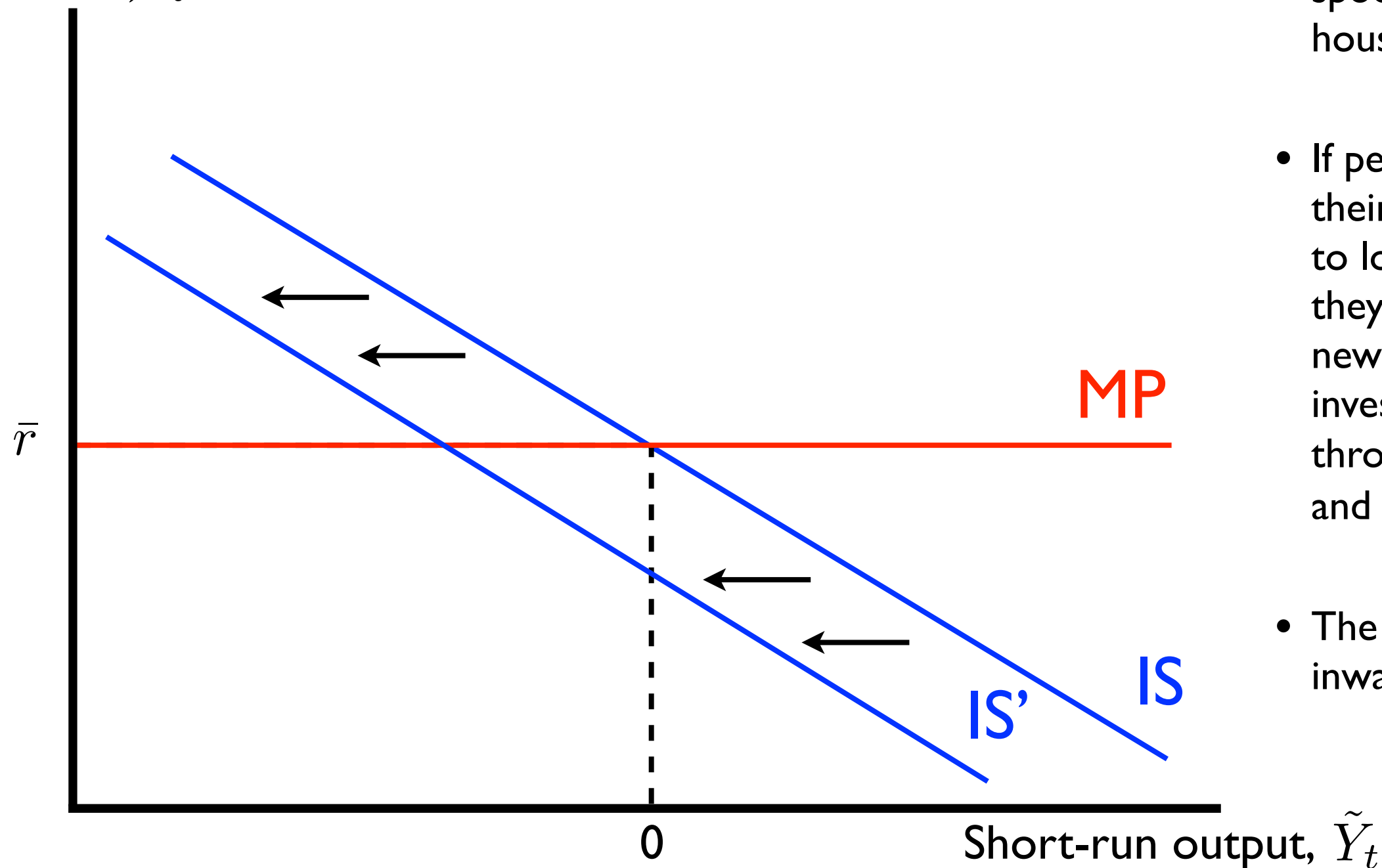
Suppose the Fed raises the real interest rate



- This is an upward shift in the MP curve
- At the higher real interest rate R' , investment is lower than before, because the real interest rate is above the marginal product of capital, $\bar{r}$
- A drop in investment pulls short-run output negative, below potential output, to $\tilde{Y}'$

Why might the Fed need to act? Consider a bursting housing bubble and aggregate demand

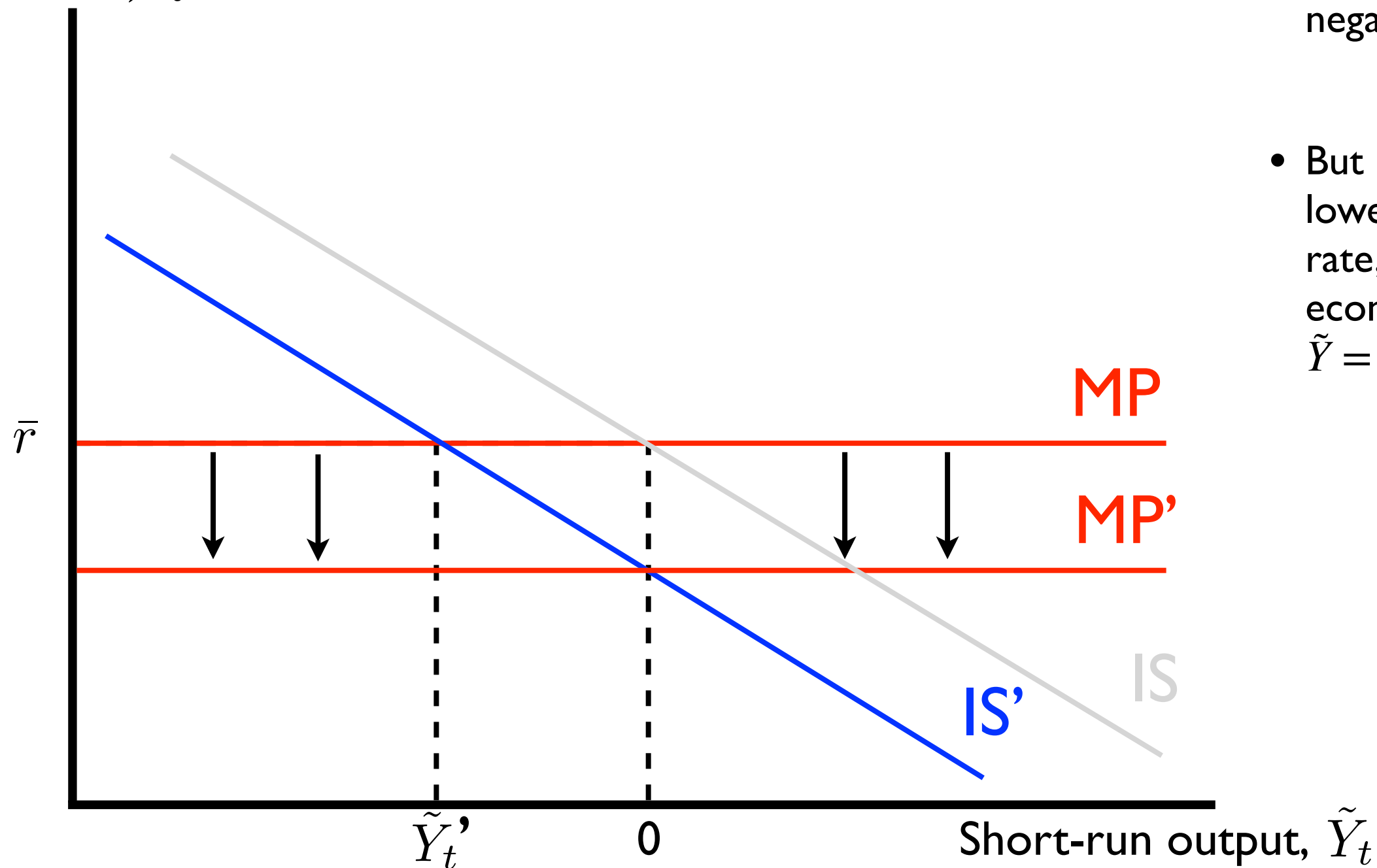
Real interest rate, R_t



- Suppose people get spooked by the housing market
- If people fear that their housing is about to lose value quickly, they might stop buying new housing, shifting investment down through a decline in $\bar{a}_i$ and in $\bar{a}$
- The IS Curve shifts inward to the left

Without action, short-run output will decline

Real interest
rate, R_t



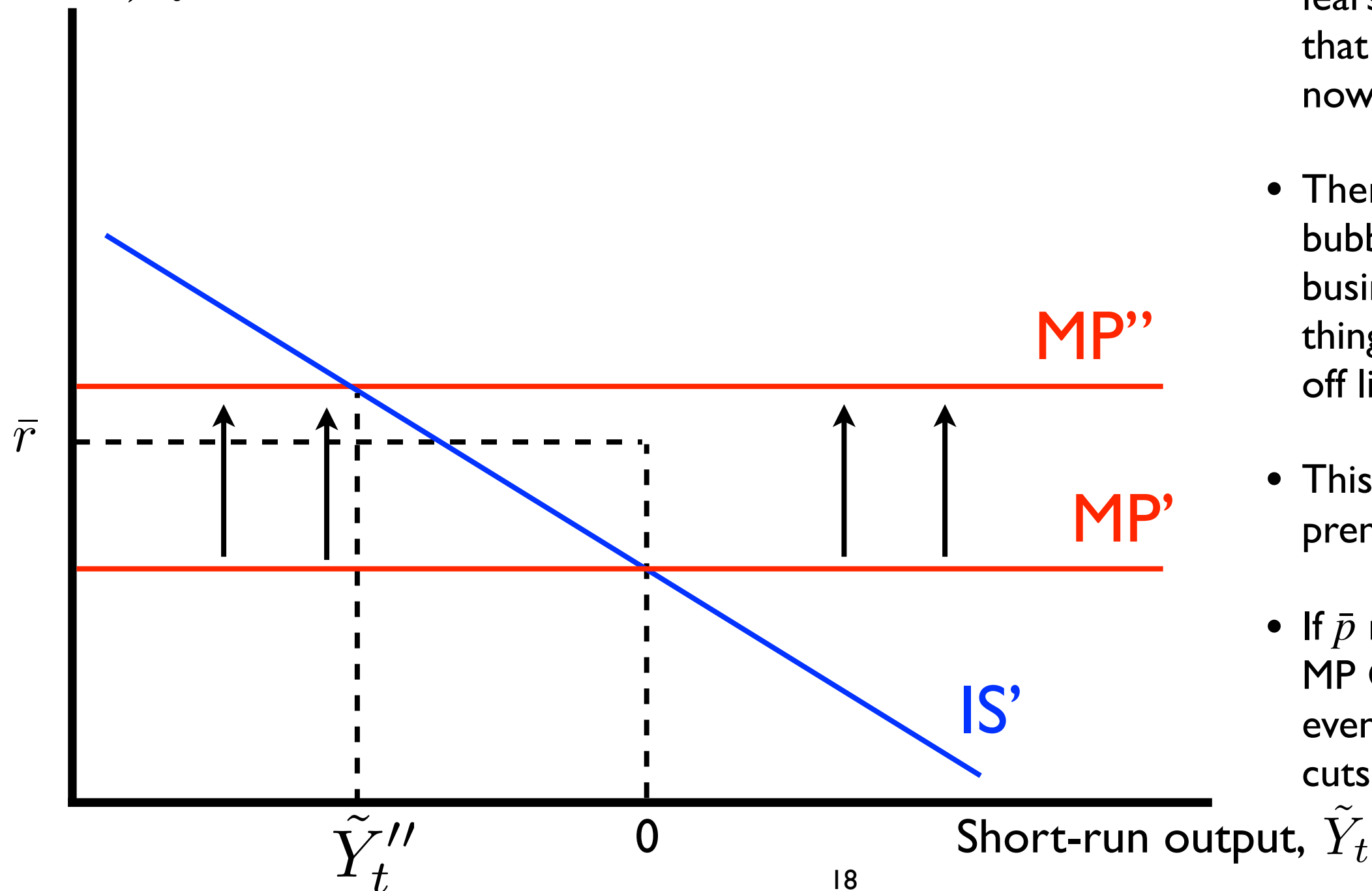
- The IS curve has fallen, to IS' , which will pull short-run output negative
- But if the Fed can lower the real interest rate, it can push the economy back to $\tilde{Y} = 0$

Perceptions of risk can create difficulties

Real interest
rate, R_t

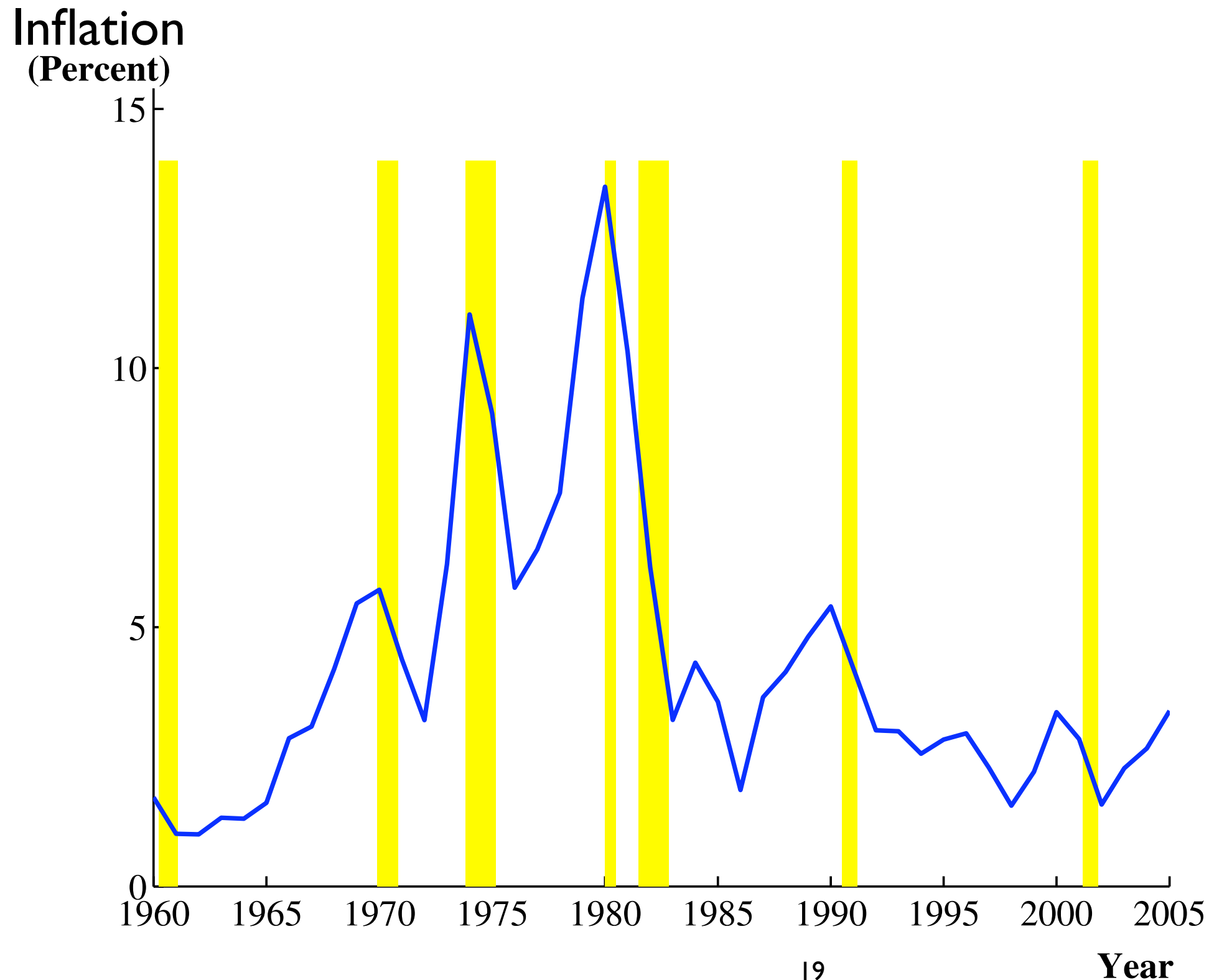
$$R_t = R^{ff} + \bar{p}$$

The MP Curve



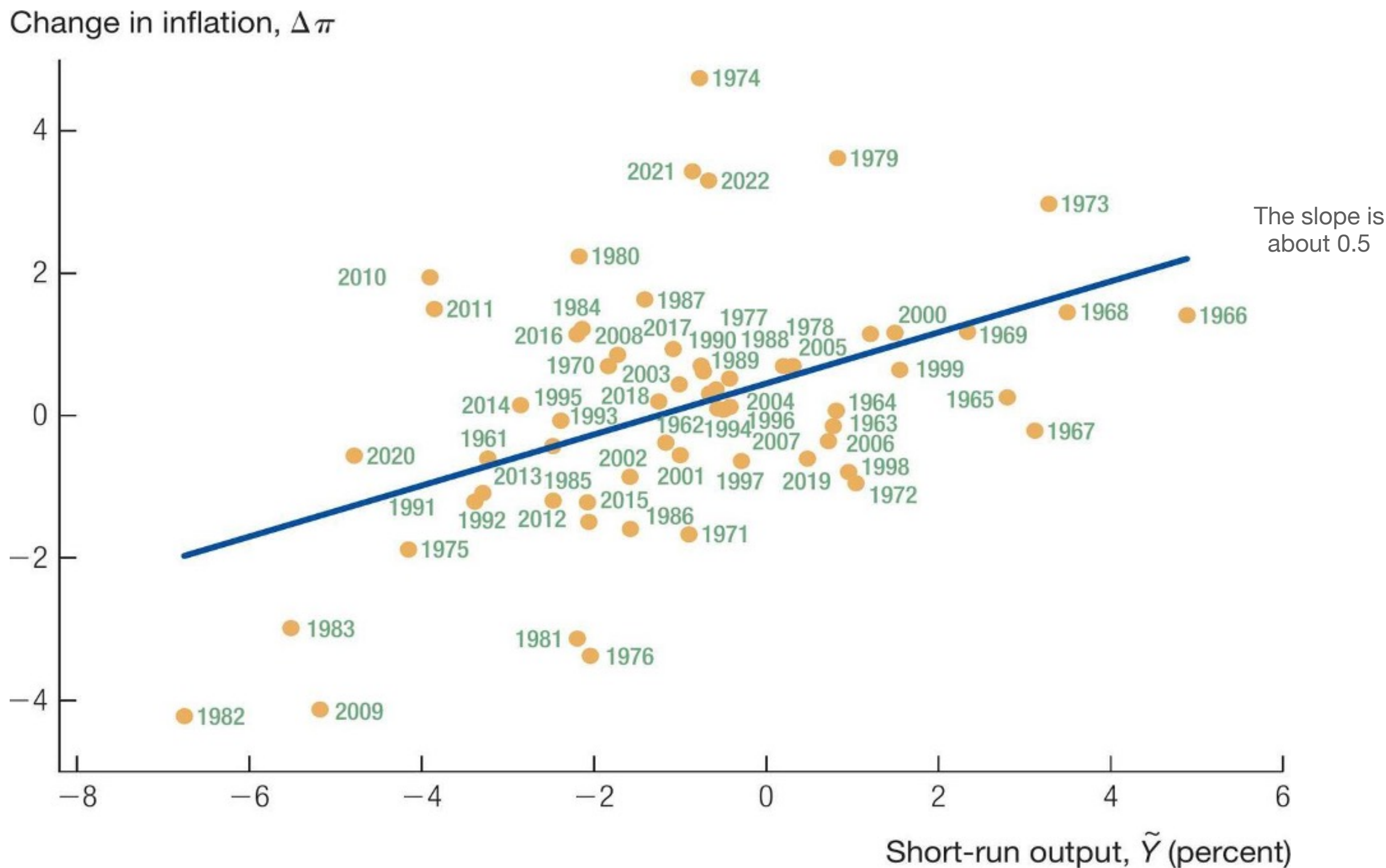
- Suppose a bursting of housing bubble raises fears among banks that their loans are now riskier
- There was an asset bubble, which means businesses did dumb things that didn't pay off like they thought
- This raises the risk premium $\bar{p}$
- If $\bar{p}$ rises enough, the MP Curve *might* rise even after the Fed cuts R^{ff}

Next: We have seen how **inflation** frequently rises during expansions and almost always falls during recessions



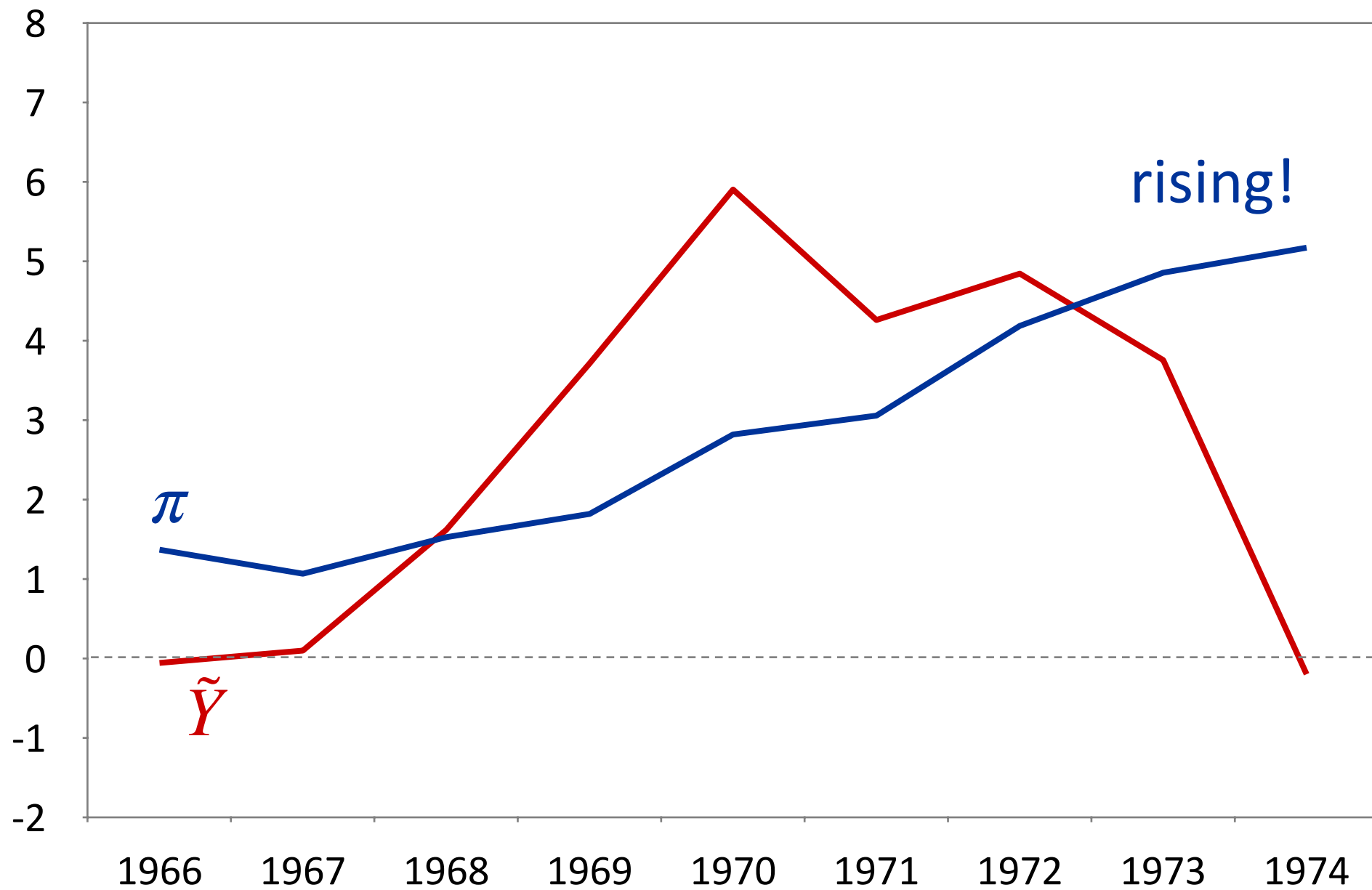
- We know that in the **long run**, inflation is tied to the growth rate of the **money supply** through the quantity theory of money: $MV = PY$
- But inflation in the **short run** is associated with booms and busts

The Phillips curve is the upward-sloping relationship between the change in inflation and short-run output



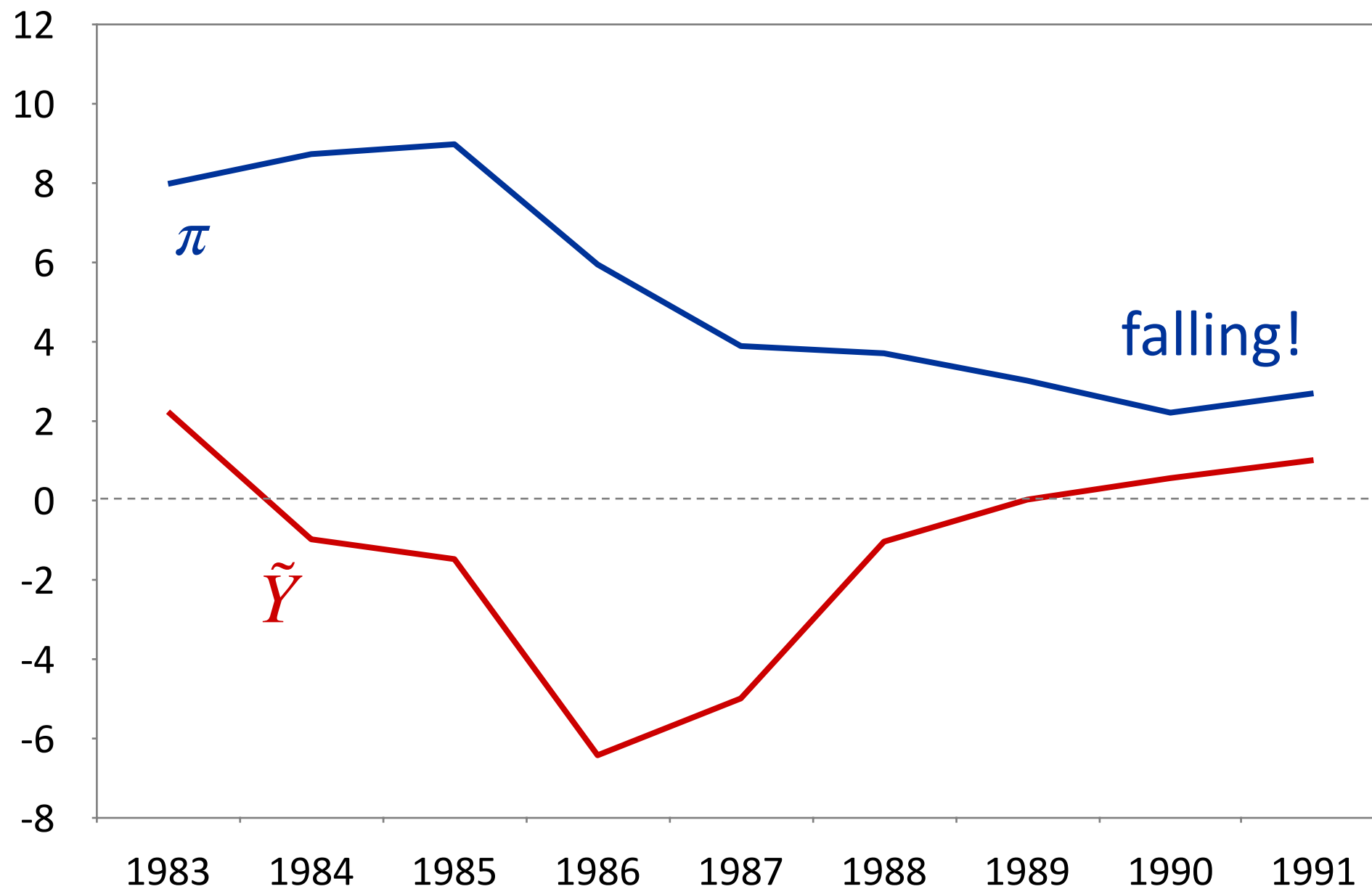
Source: FRED. U.S. Department of Commerce.

Inflation and Output, 1962–1970



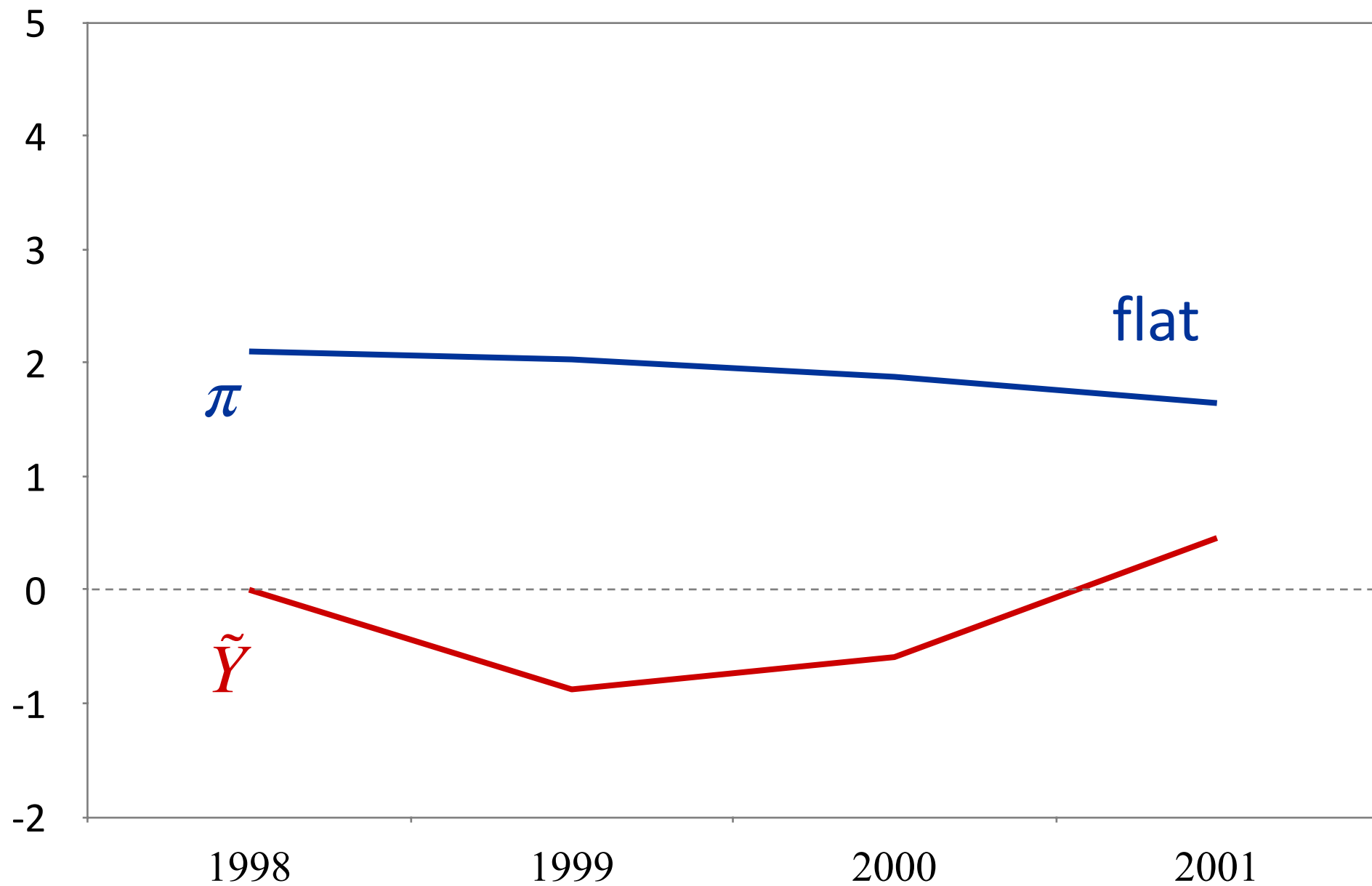
Source: Bureau of Economic Analysis.

Inflation and Output, 1979–1987



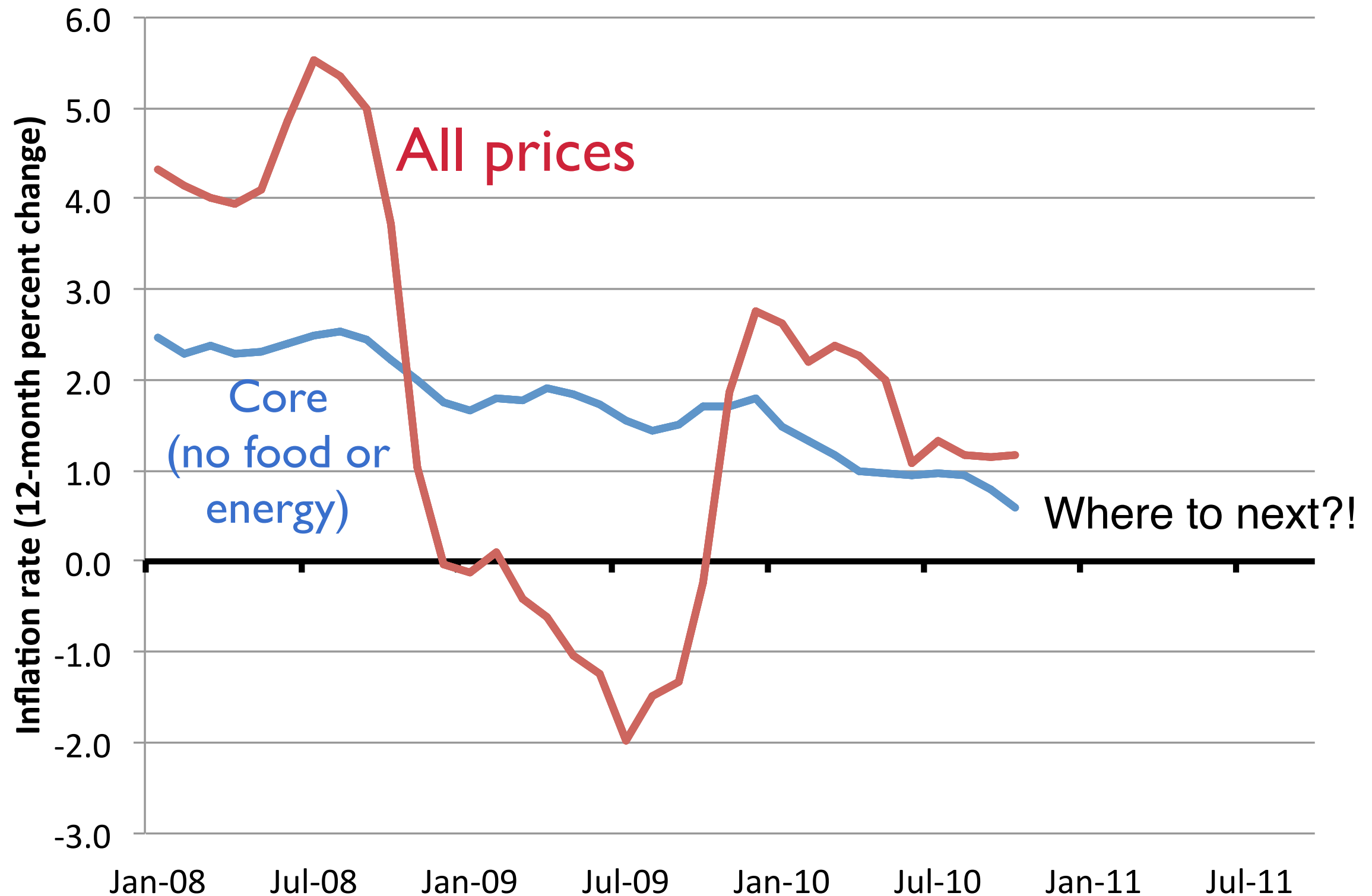
Source: Bureau of Economic Analysis.

Inflation and Output, 1994–1997

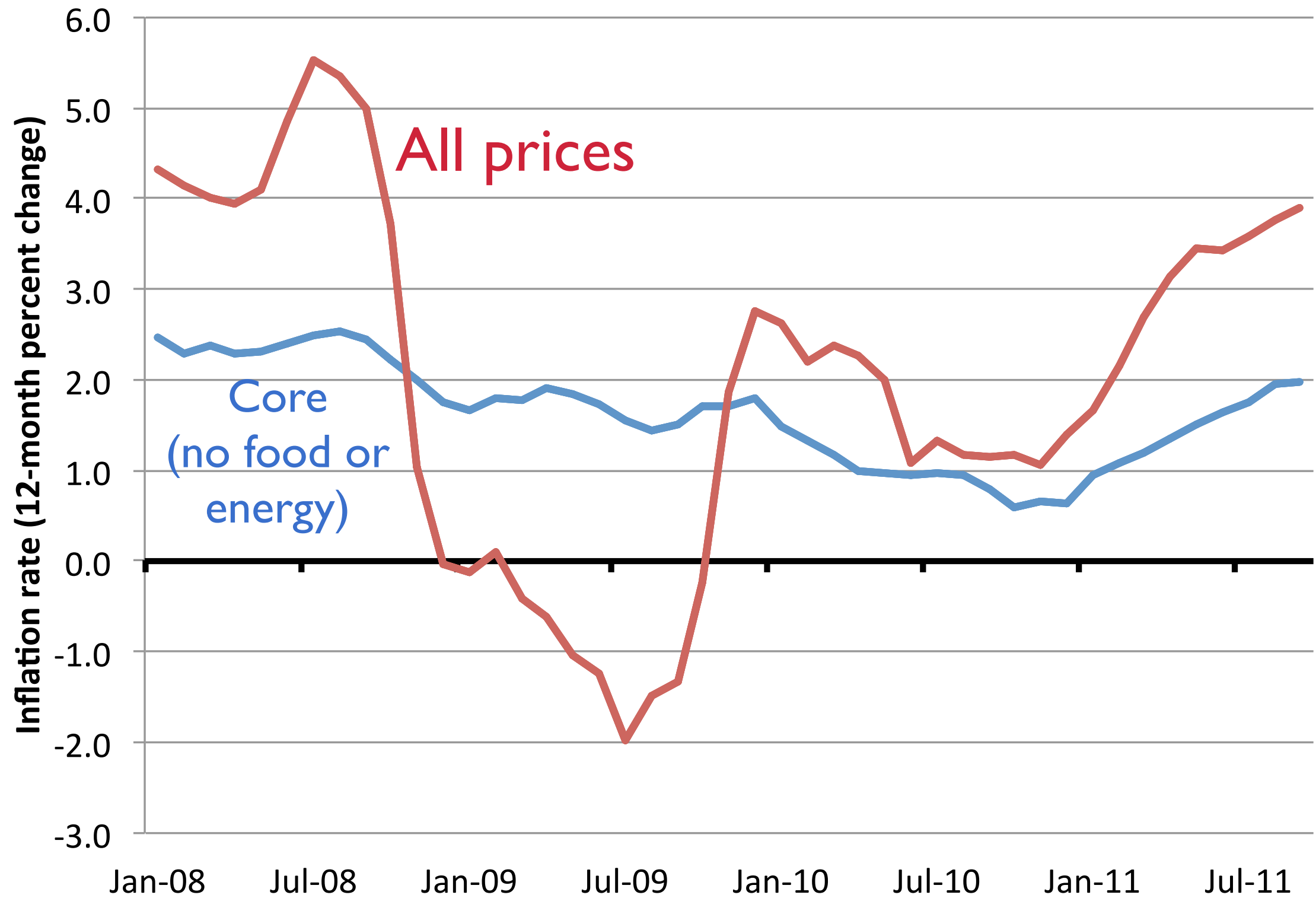


Source: Bureau of Economic Analysis.

The Great Recession I: Falling, even negative inflation?



The Great Recession II: Rising but moderate core inflation



How does inflation happen?

Firms set their prices

- The money supply *must* determine the price level in the long run, but in the short run, the action is with firms and prices
- Imagine a store that produces and sells ice cream
- For the last several years, it has noticed that all prices are rising at 3% each year, so $\pi = 0.03$ usually
- But suppose the owner of the ice cream store reads in the paper that many local workers just lost their jobs — and the local workers are the folks who usually eat ice cream
- The owner might lay off some workers and supply less ice cream, but he/she will probably raise prices less than 3%

What does this price-setting behavior mean for our model?

- Actual economy-wide inflation will be equal to what the firm owners *think* economy-wide inflation will be ...
 - Minus something when owners expect a slump, or
 - Plus something when owners expect a boom

$$\begin{array}{ccccc} \text{Actual inflation} & \nearrow & \pi_t = & \pi_t^e & + \bar{v}\tilde{Y}_t & \nwarrow & \text{an increasing} \\ & & & \uparrow & & & \text{function of} \\ & & & \text{Expected inflation} & & & \text{short-run output} \end{array}$$

Suppose $\bar{v} = 0.5$ and $\pi_t^e = 0.03$. Then when output is 1% above potential, $\tilde{Y}_t = 0.01$ and inflation will be higher too:

$$\pi_t = 0.03 + 0.5(0.01) = 0.035$$

But what are expectations about inflation — π_t^e ?

- Sometimes they can really get out of control, which they did in the U.S. during the 1970s
- If firms and consumers expect high inflation, then it is likely to happen if agents are rational

$$\pi_t = \pi_t^e + \bar{v}\tilde{Y}_t$$

- But suppose inflation is relatively subdued. Then firms will expect inflation to be what it has been in the recent past:

$$\pi_t^e = \pi_{t-1}$$

- Under these “adaptive expectations,” we find that

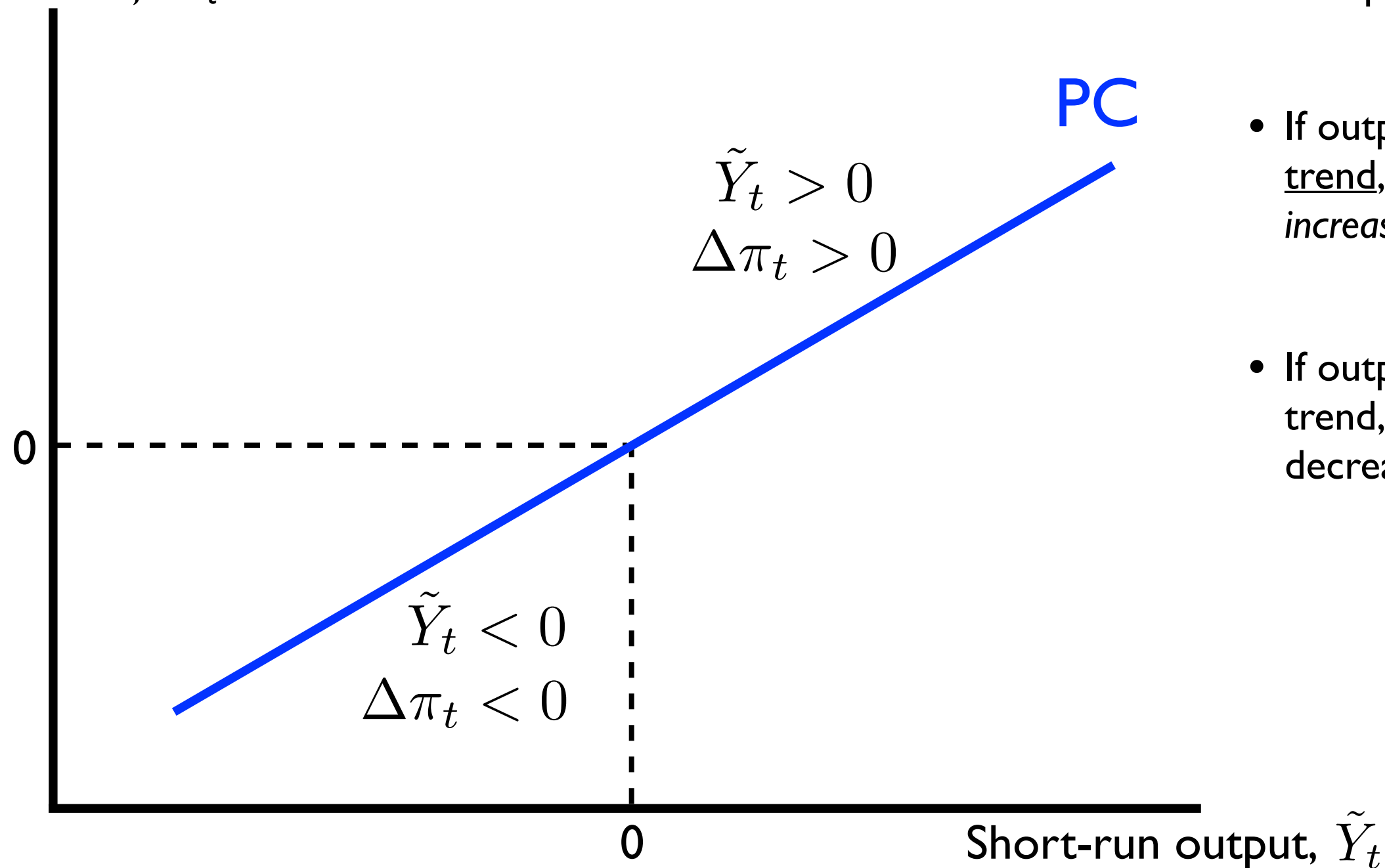
$$\pi_t = \pi_{t-1} + \bar{v}\tilde{Y}_t \quad \text{so}$$

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

This equation is the Phillips Curve

$$\Delta\pi_t = \bar{v}\tilde{Y}_t$$

Change in
inflation, $\Delta\pi_t$



- We'll call this "PC" for Phillips Curve

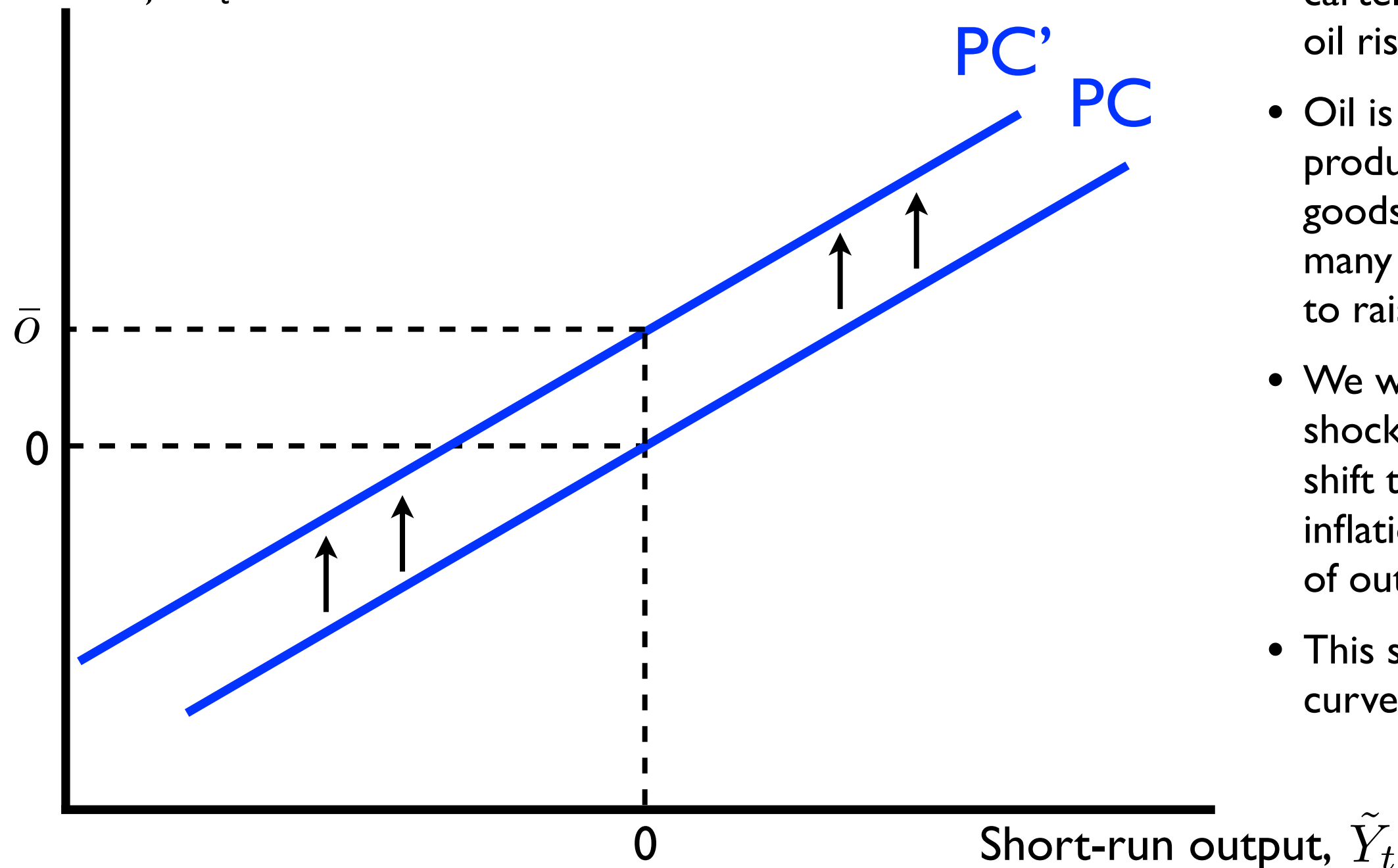
- If output is above trend, then inflation *increases*

- If output is below trend, then inflation *decreases*

Oil shocks and other events can raise inflation

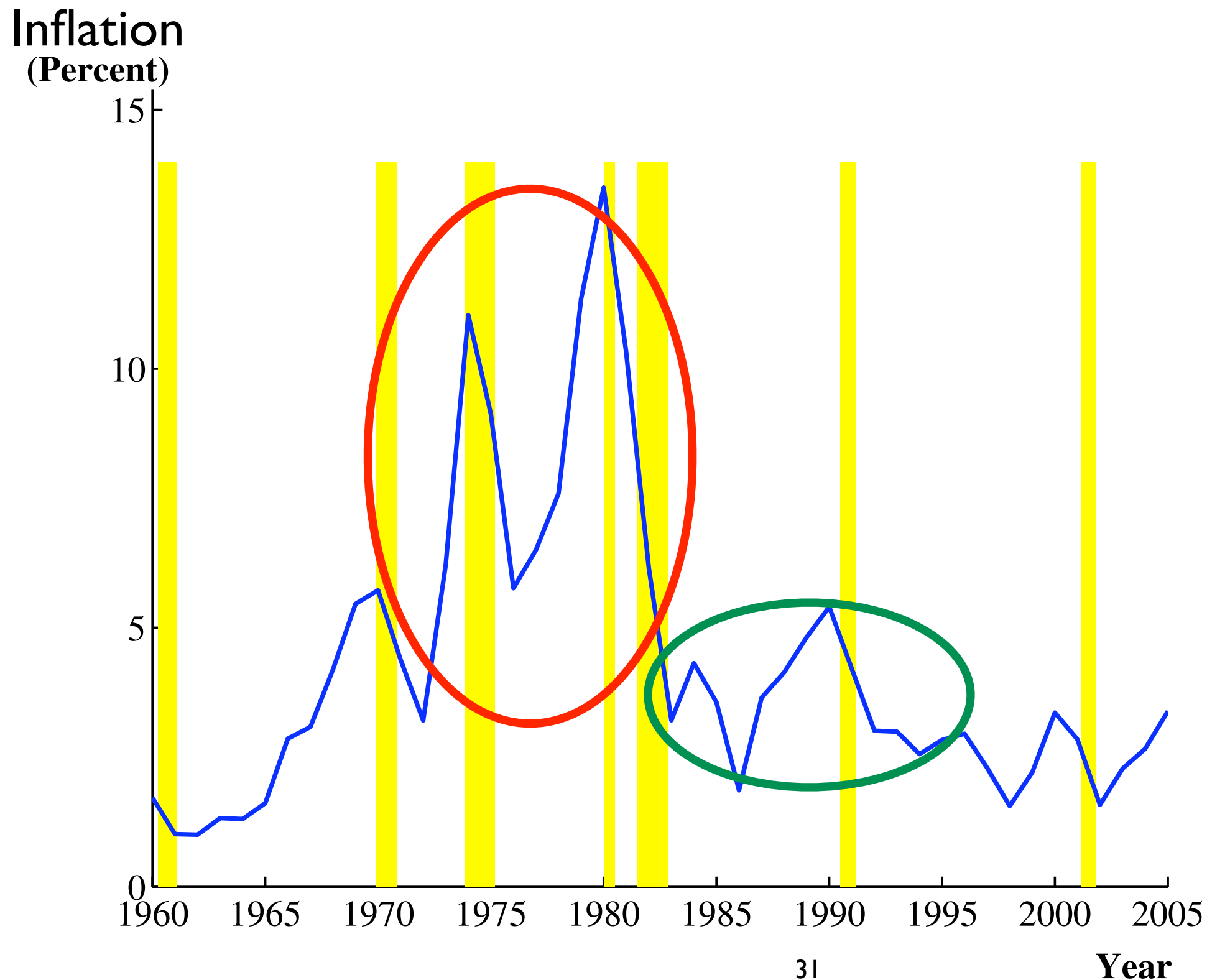
$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

Change in
inflation, $\Delta\pi_t$



- Suppose the supply of oil is restricted by the OPEC production cartel, and the price of oil rises
- Oil is an input in the production of many goods and services, so many producers have to raise prices
- We will model the oil shock as an additive shift to $\Delta\pi$, raising inflation at every level of output
- This shifts the PC curve upward

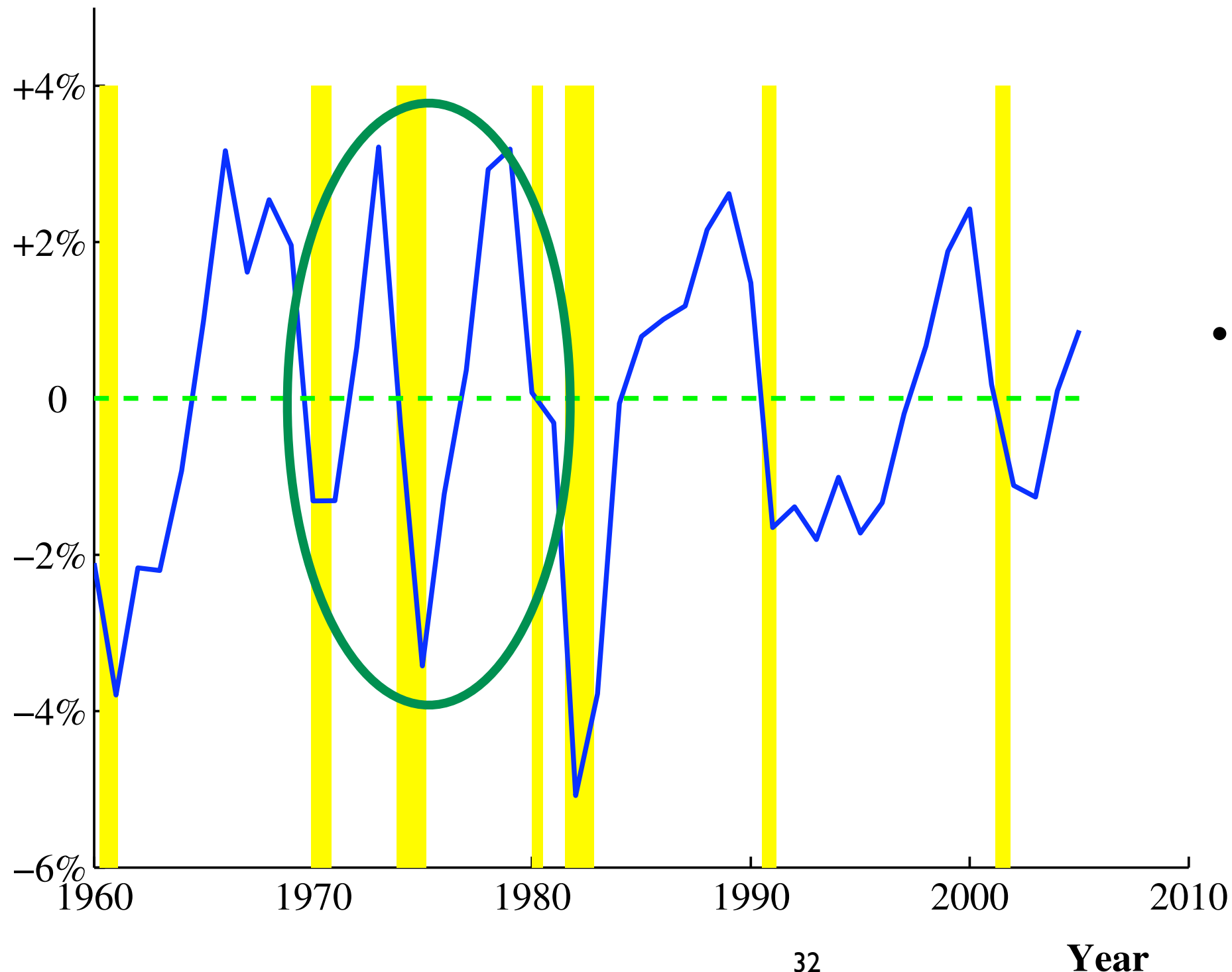
We'll use the full short-run model to explain macroeconomic history in the 1970s and 1980s



- **First**, why did inflation get to be such a problem in the 1970s, especially by 1980?
- **Second**, how did policymakers regain control over inflation and bring it back down to levels not seen since the 1960s?

Why did inflation increase in the 1970s?

Short-Run Output, $\tilde{Y}$



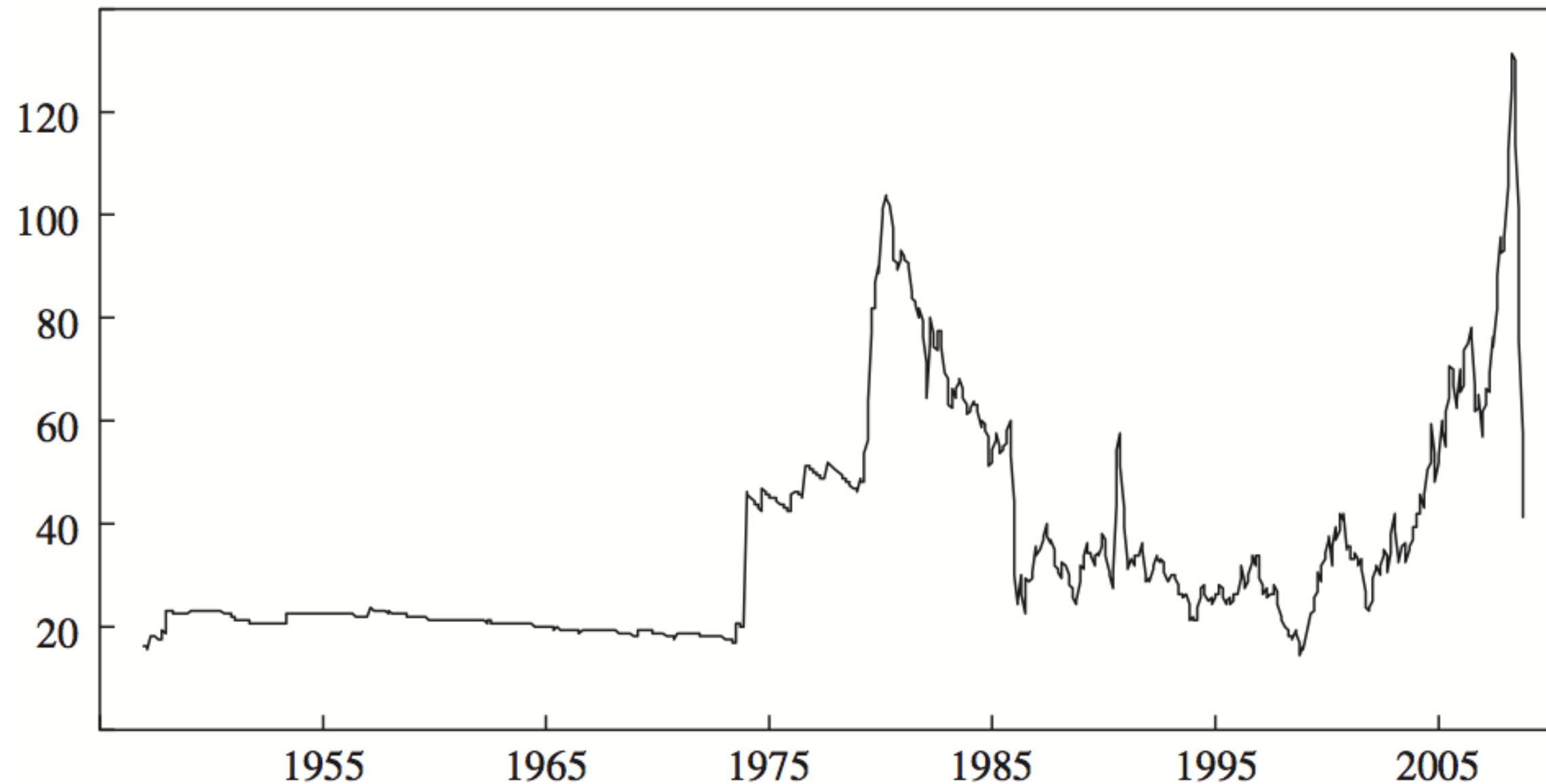
- Short-run output was swinging back and forth in a pretty normal fashion — there really wasn't a massive inflationary expansion
- So the high inflation was **not** the result of any movement along the Phillips Curve, because there really wasn't any permanent change in short-run output

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$

Per [Hamilton \(BPEA, 2009\)](#), 1970s oil shocks were big

Figure 1. Real Crude Oil Prices, January 1947 to December 2008^a

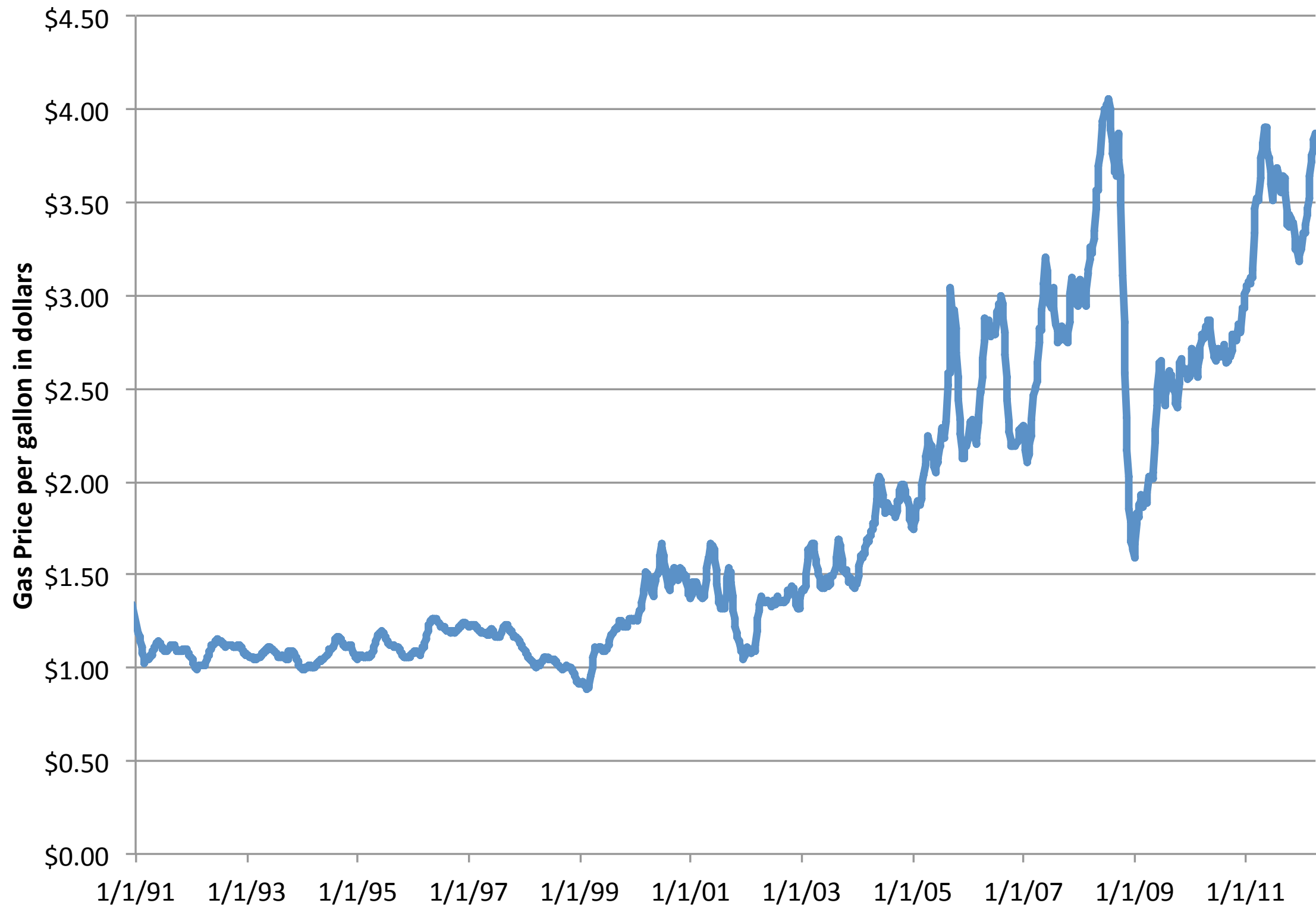
Constant dollars per barrel



Sources: Energy Information Administration (EIA) data; Bureau of Labor Statistics data.

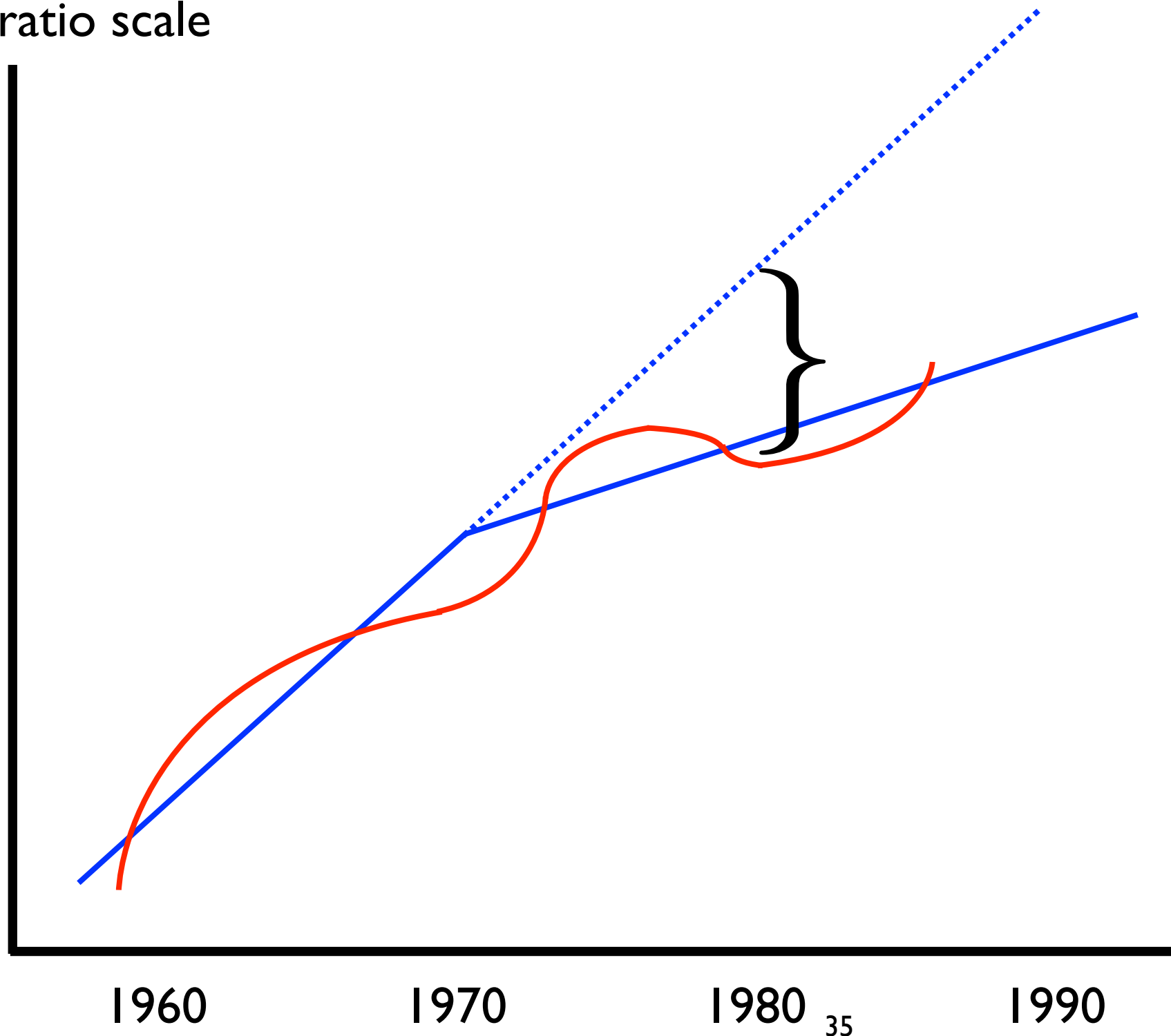
a. Monthly average price of West Texas Intermediate crude in dollars of November 2008.

Gas Prices: Do you remember 2008?



And something else happened

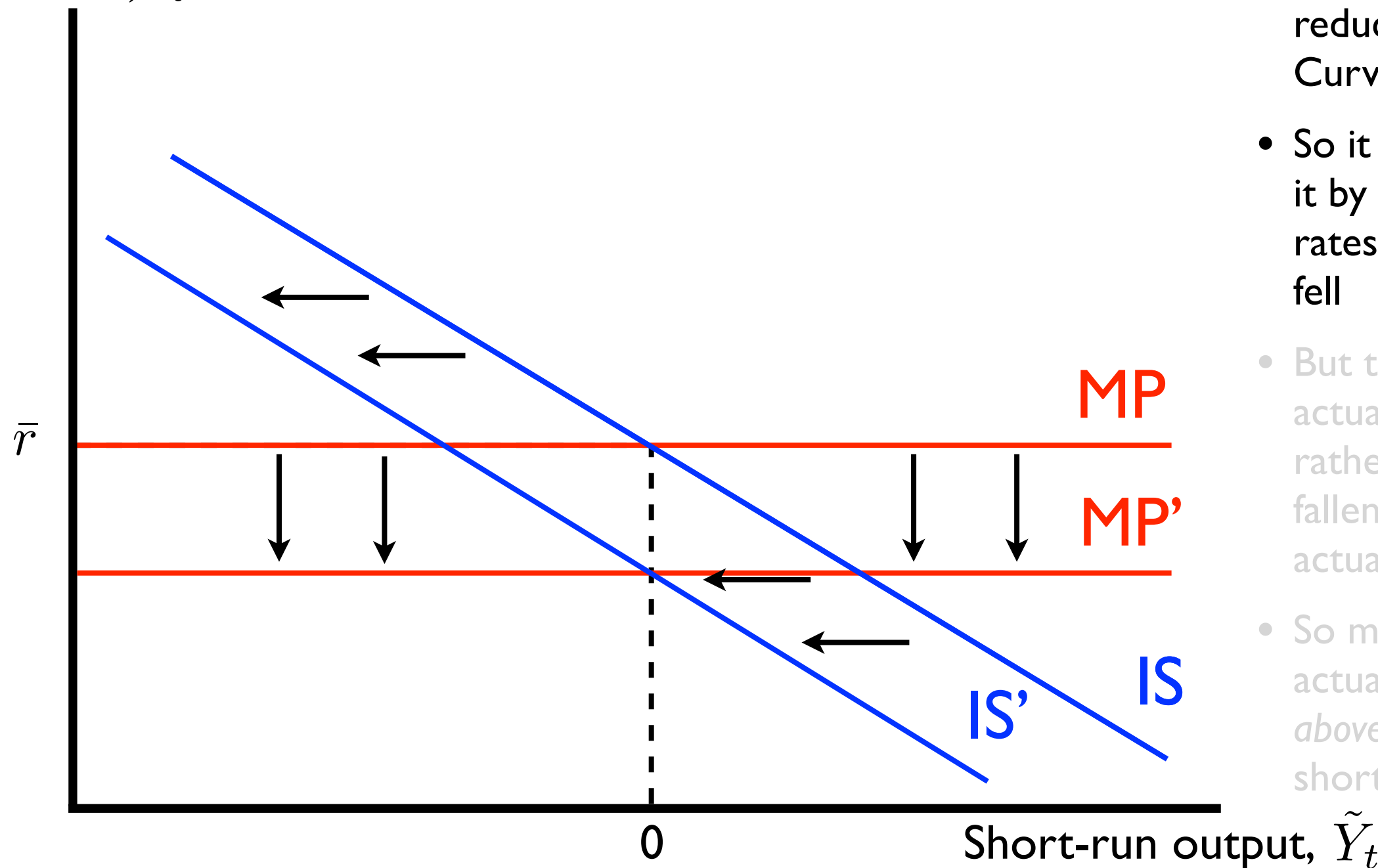
Real output,
ratio scale



- **Productivity growth slowed**, and so did growth in potential output!
- Visually, growth in potential was rapid from 1960–1970,
- And then a pretty big slowdown hit!
- We **know** this with 20/20 hindsight, but at the time, policymakers thought potential was continuing to grow
- Actual GDP was wiggling around,
- And relative to the old trend, this looked like a big recession!

Seeing what looked like a big recession, the Fed “loosened” monetary policy in response

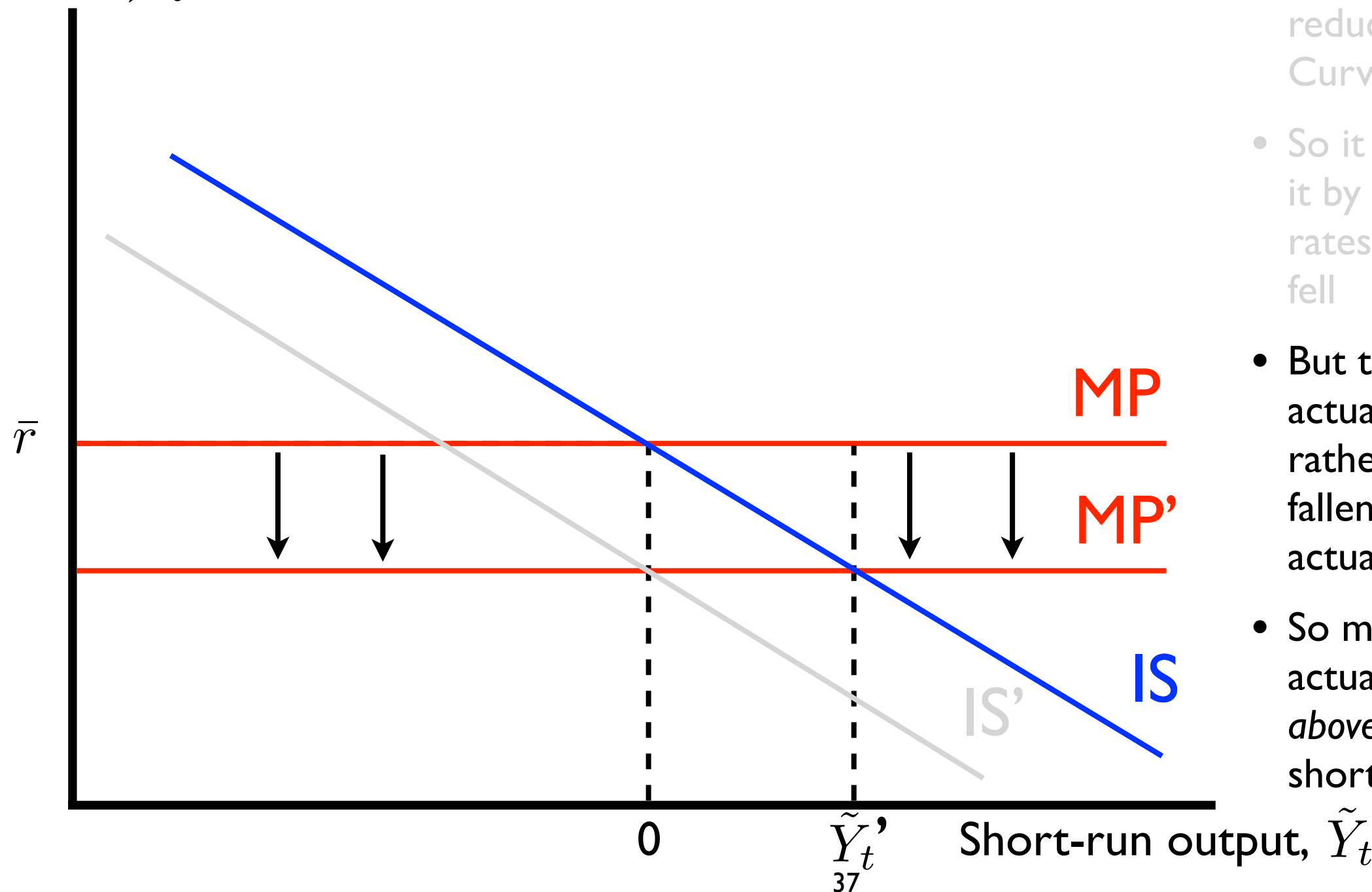
Real interest rate, R_t



- The Fed thought it saw a big recession, a reduction in the IS Curve
- So it tried to counteract it by lowering interest rates, and the MP curve fell
- But the economy wasn't actually in a recession; rather, potential has fallen. The IS Curve actually never moved
- So monetary policy is actually pushing output *above potential* in the short run!

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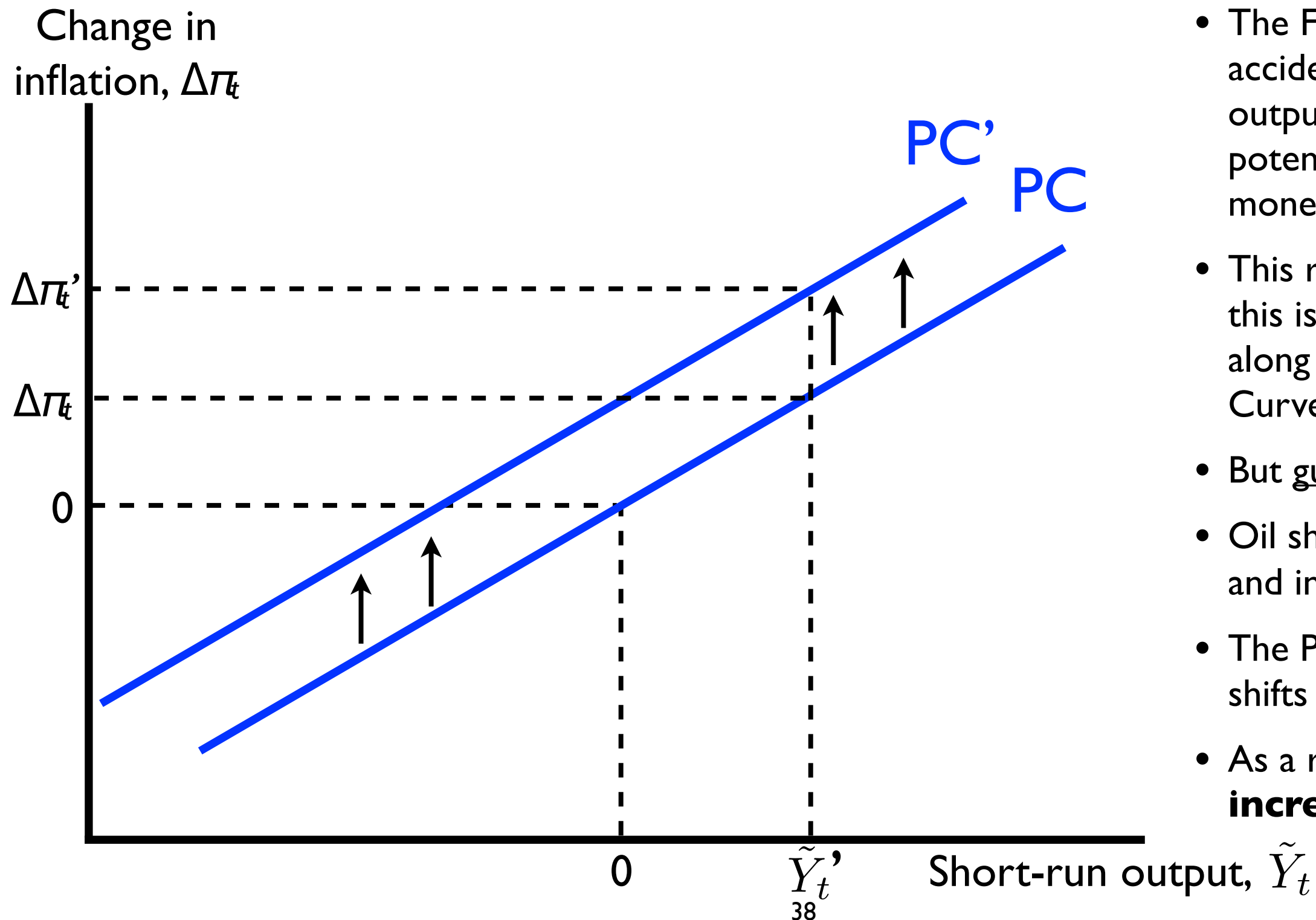
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This would be movement along the Phillips Curve

$$\Delta\pi_t = \bar{v}\tilde{Y}_t + \bar{o}$$



- The Fed has accidentally stimulated output above potential with “easy” money
- This raises inflation; this is movement along the Phillips Curve
- But guess what else?
- Oil shocks also hit and increase $\bar{o}$
- The Phillips Curve shifts upward as well
- As a result, inflation **increases a lot**

What is the result of all this?

- This is typically called the period of “stagflation” — meaning **stagnant** growth in income per capita in addition to relatively rapid **inflation**
- Again, why did this happen? Several reasons:
 - There were oil shocks that raised prices of oil and other things
 - There was a reduction in growth of potential output, and the Fed mistook this for a recession, to be grown out of, and ran “easy” money policy with lower interest rates, stimulating the economy
 - Even if the Fed had perceived things correctly, they might not have believed they needed to trigger a *temporary* recession in order to reduce inflation from the oil shocks